

Equidistribution of Integer Multiples

GREEN-052

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $c > 0$ and let A be a set of n distinct integers. Does there exist θ such that no interval of length $\frac{1}{n}$ in $\mathbb{R}/\mathbb{Z}$ contains more than n^c of the numbers $\theta a \pmod{1}$, for $a \in A$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>